Project Reflection.

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Project: Knowledge distillation and deployment of a lightweight command-parsing language model.
Github: https://github.com/Abhimanyu-Space/cloud-command-parse-distillation.

In this assignment, I worked on building a lightweight command-parsing language model using knowledge distillation.

The main objective was to use a large teacher model to generate structured command data and train a smaller student model that could run locally.

Implemented teacher-based data labelling using Phi-3.5, trained a Pythia-70M student model, and developed a FastAPI service for command parsing. I also worked with Docker, kubernets, new access control rules and Locust-based load testing.

But, during this, I have faced many challenges, Some teacher-generated response contained missing fields, invalid intents or additional fields, so they require validation and filtering. Training the student model also required significant time & computational resources.

The evaluation shows 100% valid JSON output, as (I changed the "null" to null). but the full-parse accuracy was only 50%. This indicates the producing valid JSON does not necessarily mean that all extracted command information is correct.

Some assignment requirements are still incomplete, including the specified 1,500-record training limit, object storage integration, complete API endpoints, Kubernetes config, and CI/CD automation.

This assignment helped me understand that model deployment involves much more than training a model. Data quality, evaluation, API reliability, security, and infrastructure are equally important. In future, I would focus on completing the missing requirements and improving the model's command extraction accuracy.